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(1) Research Program/Project/Study Title: Numbers and Fortitude: A Grounded Theory on Mathematical Grit in Filipino Students	
Proponent Name:	Alvin D. Tenorio
Academic Rank:	Asst. Prof. IV
Co-Authors (with academic rank):	Liezyl Igdanes – Instructor I John Patrick Eswan – Contractual Instructor
Campus/Satellite College/Address:	CapSU Pontevedra Campus – Bailan, Pontevedra, Capiz
Telephone/Cell Phone/Fax Number:	09566907315
E-mail Address:	adtenorio@capsu.edu.ph

Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Pontevedra
Province:	Capiz
Region:	Region VI

(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☒ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
Research:	Development:						
☐ Basic	☐ Pilot Testing						
☒ Applied	☐ Tech. Promotion/Commercialization						
(7) Sector/Commodity Education	(8) Discipline/Field of Specialization Psychology, Education, Mathematics						

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(9) Rationale

Intelligence was once thought to be critical in determining which students would succeed in school and which would fail (Allen & Bond, 2001; Kidd & Latif, 2003). It was not until the 1970s that psychologists challenged this notion and discovered that academic achievement extends beyond IQ (Poropat, 2009).

Angela Duckworth's groundbreaking study established grit as a psychological trait that significantly predicts academic achievement, surpassing conventional metrics like IQ or talent (Duckworth et al., 2007). Grit research has predominantly leaned on quantitative methodologies, particularly in crafting and validating grit scales (Duckworth et al., 2009; Duckworth & Quinn, 2009). Subsequent studies reinforced these findings by highlighting robust correlations between grit and academic outcomes, self-regulation, and mathematics performance (Credé et al., 2017; Wolters et al., 2017; Duckworth & Eskreis-Winkler, 2013; Turner & Patrick, 2004). Recent scholarly pursuits, exemplified by Rostiga et al.'s (2021) exploration of academic grit, have broadened the research scope.

However, a noticeable gap persists in the literature regarding the explicit investigation of grit formation among Filipino students. Despite the burgeoning emphasis on grit in education, a comprehensive understanding and empirical investigation of the development and manifestation of mathematical grit among Filipino learners remains conspicuously absent. The distinctive cultural context and challenges encountered by Filipino students in mathematics learning underscore the necessity for a dedicated inquiry into the nuanced complexities of grit within this context. The insights that will be garnered from this study are poised to contribute significantly to a comprehensive model elucidating the manifestation and cultivation of grit among diverse Filipino learners.

This study aims to develop a theory that could explain the multifaceted nature of mathematical grit among Filipino learners by investigating the dimensions perceived by the learners. Additionally, it seeks to analyze the diverse factors influencing the development and expression of mathematical grit among Filipino learners, exploring the impact of these factors on their overall grit development. Moreover, it aims to delve into how the perception of challenges, setbacks, and failures in mathematical learning shapes the cultivation of grit among Filipino learners.

(10) Statement of the Problem

Generally, this study aims to develop a theory that could explain the underlying components and processes that contribute to the development and manifestation of mathematical grit among Filipino learners. These are the subquestions of the study:

1. What dimensions or facets constitute mathematical grit as perceived by Filipino learners?
2. What factors influence the development and expression of mathematical grit among Filipino students? How do these factors impact this development?
3. How do Filipino learners perceive challenges, setbacks, and failures in mathematical learning and how does this perception influence the development of grit?

(11) Review of Literature

Research on Grit

Research, primarily led by Duckworth, emphasizes that students with higher levels of grit are more likely to surmount obstacles, adapt to difficulties, and achieve superior learning outcomes. The meta-analysis by Credé et al. (2017) confirms a positive relationship between grit and achievement outcomes across diverse settings. Grittier individuals consistently achieve higher academically compared to their counterparts. Studies by Datu et al. (2018) and Lee and Sohn (2017) highlight that grittier students are more likely to be engaged in their academic studies, leading to increased academic achievement. Pate et al. (2017) found that students achieving lower grades had reduced levels of grit, reinforcing the connection between grit and academic performance. Oriol et al. (2017) emphasize that grittier students exhibit a higher sense of self-efficacy, and their perseverance in academic pursuits contributes to an enhanced belief in their own abilities, increasing their chances of academic success.

Grit entails the persistent and vigorous pursuit of challenging goals, maintaining effort, sustaining unwavering interest over extended periods despite setbacks, failures, and obstacles, as well as

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navigating phases of stability in the course of progress. Notably, grit has proven to be a valid predictor of academic success, surpassing traditional measures in efficacy (Strayhorn, 2014). Angela Duckworth's series of studies demonstrated that grit predicts successful academic outcomes, including higher GPAs and better retention in military settings (Duckworth et al., 2007).

Beyond academic success, grit has shown its predictive power in academically inclined competitions such as the National Spelling Bee (Duckworth & Quinn, 2009; Duckworth et al., 2011). Grit remains instrumental in predicting academic achievement, even when considering past academic performance and aspirations (Strayhorn, 2013). Gritty individuals, in contrast to those low in grit, exhibit stamina, persistently staying on track to attain their goals, while individuals lacking grit may easily become disheartened or bored, abandoning effort or opting for alternative actions (Duckworth et al., 2007).

Recognized as a personal quality common in leaders, grit emerges as a crucial antecedent of success and excellence in every domain, irrespective of giftedness or talent (Duckworth et al., 2007; Duckworth & Yeager, 2015). Eskreis-Winkler and colleagues (2014) demonstrated that grit significantly predicts academic performance, surpassing the predictive validity of standardized test scores. Similarly, Credé et al. (2016) underscored the role of grit in predicting academic achievement across various educational levels.

A study by Eskreis-Winkler et al. (2014) found that perseverance is associated with higher academic achievement and positively influences students' ability to navigate the complexities of educational environments. Additionally, passion, characterized by intense and enduring interest in one's chosen pursuits, is another critical dimension of grit (Duckworth et al., 2007). Research by Vallerand et al. (2003) suggests that individuals who are intrinsically motivated and experience high levels of passion for their activities are more likely to demonstrate sustained effort over time. Furthermore, grit has been linked to positive outcomes in various domains, including education, career success, and personal well-being (Duckworth et al., 2007; Eskreis-Winkler et al., 2014).

Recent studies have explored the role of grit in various domains, such as education, business, and sports. Eskreis-Winkler et al. (2014) found that grit predicts academic achievement better than traditional measures such as intelligence quotient (IQ), highlighting its crucial role in educational success. In the business realm, grit has been associated with higher levels of performance and career advancement (Muenks et al., 2017). Moreover, in sports, athletes with high levels of grit demonstrate enhanced perseverance, contributing to improved athletic performance (St Clair-Thompson & Gathercole, 2006).

Grit in Mathematics Education

Mathematical grit, within the context of mathematical learning and problem-solving, can be defined as unwavering perseverance and passion for long-term goals. This concept encapsulates a student's ability to persist through challenging mathematical tasks, rebound from failures, and maintain a positive attitude toward mathematics. Derived from Angela Duckworth's broader notion of grit, which underscores the importance of perseverance and passion in achieving long-term success, mathematical grit has garnered attention in the realm of mathematics education.

Research by Beal et al. (2019) delved into the relationship between grit and mathematics performance, revealing that students with higher levels of grit tend to excel in mathematics. Moreover, grit was identified as a positive influence on mathematics self-efficacy, as indicated by the work of Muenks et al. (2017), emphasizing its potential significance in the context of mathematical learning. While not explicitly centered on grit, Dweck's mindset research aligns with the role of grit in mathematics education, highlighting the importance of fostering a growth mindset to enhance perseverance and achievement in challenging subjects like mathematics (Dweck, 2006).

Grit has also been linked to mathematical problem-solving, a fundamental aspect of mathematical proficiency. Gustems-Carnicer et al. (2019) found that grittier students exhibit greater tenacity and resilience when faced with complex mathematical challenges. Paunesku et al. (2015) demonstrated that grit enhances mathematics self-efficacy, positioning it as a crucial factor in mathematical achievement. They also highlighted the positive impact of fostering a growth mindset on student performance in mathematics by reducing anxiety and promoting grit.

Studies by Duckworth and Eskreis-Winkler (2013) and Credé et al. (2016) emphasized the role of grit in determining students' engagement, motivation, and learning outcomes, particularly in challenging

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subjects like mathematics. Turner and Patrick (2004) explored the relationship between grit and mathematical performance among undergraduates, concluding that higher levels of grit were associated with better academic outcomes in mathematics. Sandilos et al. (2016) investigated how grit influences students' mathematics self-efficacy and attitudes, underscoring its potential impact on shaping perceptions of mathematical abilities.

(12) Scientific Basis/Theoretical Framework

Grit Theory

Over the past decade, there has been a growing interest in discerning the factors contributing to individuals' success, particularly in education (Duckworth et al., 2007). Among the non-cognitive factors gaining increased attention, grit stands out prominently. Defined by Angela Duckworth and colleagues at the University of Pennsylvania as "the perseverance and passion for long-term goals," grit represents a performance character strength that individuals draw upon to realize their potential in the face of challenges (Duckworth et al., 2007, p. 1087).

Grit theory, as conceptualized by psychologist Angela Duckworth, posits that individual achievement and success extend beyond mere talent or intelligence. Instead, they are significantly influenced by the perseverance and passion individuals demonstrate towards their long-term goals. Duckworth defined grit as the combination of passion and perseverance over an extended period, asserting that this dual quality is a more reliable predictor of success than traditional measures of talent or IQ (Duckworth et al., 2007).

The theoretical framework of grit theory draws from decades of psychological research on achievement, motivation, and personality traits. A key aspect of grit theory is the concept of passion, aligning with Csikszentmihalyi's Flow Theory suggesting that individuals are more likely to achieve optimal experience and performance when deeply engaged in activities they are passionate about (Csikszentmihalyi, 1990). Research on goal-setting theory, particularly Locke and Latham's work, supports the idea that setting specific and challenging goals, coupled with sustained effort, leads to higher levels of performance and achievement (Locke & Latham, 2002). Duckworth's grit theory also incorporates elements from Dweck's mindset theory, emphasizing the importance of a growth mindset in fostering grit by encouraging individuals to view challenges as opportunities for learning and improvement (Dweck, 2006).

Grit theory aligns with the broader perspective of positive psychology, emphasizing strengths and virtues that enable individuals to thrive (Seligman & Csikszentmihalyi, 2000). The theoretical framework of grit theory integrates insights from various psychological perspectives to provide a comprehensive understanding of how passion and perseverance contribute to individual success and achievement over time.

Grounded Theory

Grounded Theory, originally developed by Glaser and Strauss in the 1960s, is a qualitative research approach that aims at developing theories from empirical data rather than testing pre-existing hypotheses. Its theoretical framework is founded on the philosophical foundations of symbolic interactionism, phenomenology, and pragmatism (Charmaz, 2006; Glaser & Strauss 1967). This approach focuses on comprehending social phenomena from the participants' perspectives, taking into account their experiences and interactions, and its key tenets include constant comparative analysis, theoretical sampling, and theoretical sensitivity. Constant comparative analysis involves comparing data during collection and analysis to assist in developing concepts and categories. In their 1967 paper, Glaser and Strauss proposed theoretical sampling, which involves the intentional selection of participants and data sources based on emerging theoretical insights, thereby aiding in the refinement of the developing theory. Theoretical sensitivity refers to a researcher's ability to conceptualize and analyze data in a way that captures underlying patterns and structures (Charmaz, 2006). Grounded Theory, as a methodology, has been extensively utilized across disciplines, including sociology, nursing, and management, to explore and generate theories from a wide range of social phenomena (Corbin & Strauss, 2008; Glaser, 1978; Strauss & Corbin, 1990).

In the field of education, Glaser (1978) utilized Grounded Theory to investigate the process of

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student learning, resulting in the identification of key factors influencing academic achievement. These examples illustrate the theory's versatility in capturing the multifaceted nature of human experiences and developing substantive theories alongside practical applications.

Furthermore, the theory's flexibility allows it to be employed in a variety of research contexts and paradigms. Charmaz (2006) suggested a constructivist version of grounded theory, emphasizing the researcher and participant's co-construction of knowledge. This adaptation correlates with the evolving nature of qualitative research methodologies, acknowledging the crucial role of subjectivity and interpretation in the research process. The ongoing development and refinement of Grounded Theory demonstrate its relevance and adaptability in addressing contemporary research questions and challenges across disciplines (Charmaz, 2014).

(13) Methodology

Research Design

This investigation seeks to develop a theoretical understanding of the development and manifestation of mathematical grit among Filipino learners. To achieve this objective, the decision has been made to employ a grounded theory methodology (GTM). Grounded theory methods encompass the techniques utilized for the collection and analysis of data to formulate theories (Bryant and Charmaz, 2007). These theories are derived from empirical evidence, providing analytical insight into the phenomena being examined.

The application of constructivist grounded theory enables researchers to follow the data's lead and delineate occurrences throughout the research process (Glaser & Strauss, 1967). Data tracking in constructivist grounded theory involves encoding and categorizing as the study progresses (Thornberg & Charmaz, 2014). Consequently, the collection and analysis of data using constant comparison methods will be an integral part of the process. Simultaneously, researchers will enhance data collection and analysis through memo writing to construct conceptual analyses and theoretical sampling to refine emerging theoretical ideas. Additionally, the use of constructivist grounded theory facilitates the exploration of factors and their implications for the research. This approach is deemed suitable due to its flexibility in data collection which allows for the interpretation of meaning influenced by the social context from which the data is derived (Thornberg & Charmaz, 2014).

Moreover, the use of constructivist grounded theory establishes a reciprocal relationship between researchers and participants (Chong & Yeo, 2015), positioning the researchers as contributors in the data construction process (Karpouza & Emvalotis, 2019). Consequently, the study's design aligns with the constructivist notion that knowledge is constructed through interaction (Crotty, 1998).

Context of the Study

The study will involve Grade 11 learners representing rural/urban schools, which may be considered small, medium, big, mega from regular and special classes to account for variation that may come from different contexts. These schools are considered appropriate because they are all catering to Grade 11 learners students in Capiz. They are accessed by research participants with varying socioeconomic and educational backgrounds, and from different geographic locations across the province of Capiz. Nevertheless, this research is qualitative, and accordingly, it does not seek to generalize its findings (Charmaz, 2012). This aims to offer a deep understanding of the phenomenon under study (the mathematical grit) to generate implications that could inform mathematics learning and education. Therefore, all possible secondary schools are suitable sites that can provide access to a diverse sample of information-rich participants who might have different backgrounds, views, and experiences.

Participants of the Study

Purposive Sampling. In the initial part of the data collection, purposive sampling will be used to select eligible 20-30 participants (Creswell, 2007) based on a few pre-set criteria. All participants need to be able to give informed consent to be included in the study. During the initial phase of purposive sampling emerging adult students who had satisfied the following inclusion criteria will be selected: (1)

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currently enrolled as Grade 11 SHS, (2) will not require financial compensation for their time and willing to participate.

Theoretical Sampling. Theoretical sampling will then be conducted after the first phase of data collection and analysis is complete. It will involve asking more focused questions that explain previously identified key areas of the data generated and the codes that are developed from initial transcripts. This will enable the generation of pertinent data that will illuminate different dimensions of the evolving categories and subcategories around participants' perceptions of mathematical grit. Theoretical sampling will create an opportunity to explore specific significant areas to be discovered. This will refine the researcher's interpretations of participants' views and experiences. Participants will be theoretically sampled based on their ability to provide specific information that will satisfy the dimensions of the categories to be constructed. The final sampling to be done in this sampling stage will use questions that will serve to verify and elucidate the core category whilst ensuring the breadth and explanatory power of the developing theory (Charmaz, 2008).

Participant Recruitment. The study focuses on students' understanding of mathematical grit. To do this, the researchers need to identify students who are willing to participate in the study. Their participation includes answering interviews and sharing journal entries. A consent will be secured containing their approval of participation for individual interviews, and providing journal entries they made.

Data Collection Procedures

To comprehensively explore the development and manifestation of mathematical grit among Filipino learners, a multi-faceted data collection approach will be employed. The research will utilize a semi-structured interview protocol designed to delve into participants' perceptions of mathematical grit, probing dimensions, factors influencing development, and the impact of challenges in mathematical learning. Concurrently, classroom observations will be conducted using a structured observation guide, capturing real-time instances of grit manifestation, resilience, and problem-solving approaches. Detailed field notes will be systematically documented during interviews and observations, providing contextual details, non-verbal cues, and researcher reflections for qualitative analysis.

As a crucial source of firsthand accounts, participants will be encouraged to maintain journals throughout the study period, reflecting on their experiences with mathematical challenges, setbacks, and instances of perseverance. These journals will serve as a personal narrative, offering deep insights into the participants' evolving relationship with mathematical grit. Additionally, participants will share anecdotes related to their mathematical learning journey, providing rich qualitative data that encapsulates unique perspectives and lived experiences.

Incorporating the teacher's perspective, interviews will be conducted to gain insights into observed behaviors, challenges faced by students, and the effectiveness of pedagogical strategies in fostering grit. The triangulation of data from students' journals, anecdotes, and teacher insights will contribute to a holistic understanding of the multifaceted nature of mathematical grit within diverse Filipino learning environments. The data collection procedures aim to capture both individual experiences and broader contextual factors, ensuring a nuanced exploration of grit development.

Data Analysis

The data analysis and coding procedures for this study will follow the recommendation of Charmaz (2006). The approach considers grounded theory methods as practices while emphasizing the flexibility of said practice which enables for fluid analytical approach. There will be three phases of data collection and analysis and a final verification and refining phase.

The first phase will employ purposive sampling. The first interviews aim to explore mathematics grit as perceived by Filipino learners. The interview will be done individually using interview protocol. The video will be transcribed and read multiple times before analysis that will begin with line-by-line open coding to identify the main perceptions, feelings, and issues that participants will share. There are observations to be made by the researchers using the observation guide and relevant events will be noted through field notes.

During the coding of each interview transcript, existing codes will be applied to the data. Simultaneously, new codes will be developed and compared, drawing insights from observations, field

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notes, and the reflections of students in their journals or teachers' anecdotes. This iterative process will establish links, providing a contextual framework for participants' narratives and the development of emerging analytical insights about mathematical grit.

During the second phase, the codes will be formed into groups around emergent analytical foci and tentative categories to be developed. Writing memos and making diagrams will enable the exploration of tentative categories for links within and between categories and ideas. Gaps in knowledge and properties of tentative categories are expected to be observed during this stage. The content of the interviews in the third phase aims to satisfy the knowledge gaps. The theoretical sampling will guide the direction of the study to collect data and refine the developing theory; which are the key tenets of GTM (Charmaz, 2012).

Such an iterative and cyclical approach of open and focused coding and theoretical sampling to construct and refine categories will be continued throughout the three (3) phases of data collection and analysis. The categories will gradually be developed with each rooted within the data through the interconnections and relationships with the subcategories and codes. The interviews to be conducted in the third phase will explore the topics further with limitations and establish a clear understanding of the realities to be discovered. The codes, subcategories, categories, and memos will be continuously compared, and the categories will be revised several times and refined as ideas and constructs are explored upon revisiting the interview transcripts repeatedly.

The core category and preliminary theoretical model will be outlined before the final collection of data. This leads to the verification and refining phase. In the final interview, a model will be constructed and relationships between categories will be explored. A theoretical model will continuously be refined to improve its clarity and focus through discussions and reflective memo writing.

Memos will be written throughout the data collection and analysis phases, validation, and refining stages. Different memos will be written about each of the codes, subcategories, and tentative categories, and the comparisons that will be made between participant's narratives and between codes, the similarities and differences will be observed and analyzed. Memos are to be recoded to identify similarities and differences between participants' understanding of mathematical grit.

The constant comparative method will be done by noting participants' actions and narratives and inputs from their journals, teachers' anecdotes, researchers' observation guides, and field notes to be entered into an Excel Sheet, and the characteristics and features of their perceptions and expectations along with key aspects that seemed to distinguish their views from other participants. Additionally, memos will be utilized to record and reflect on the analytical procedures to be conducted at different stages of the research to support the development from description to theoretical explanation as recommended by Charmaz (2006).

Memos vary in content based on their purpose. There are memos that will reflect of how well the interview went in relation to the interview content, dynamics, and how successfully the theoretical sampling will be accomplished. This type of memo will capture initial reflections on how the interview links to other interviews and the codes and categories. Memos will be written to explore developing categories and theoretical constructs included in the narratives supporting the analysis behind the construct and aspects of relevant literature. Making diagrams will be an essential element of conceptual memos that will be used to develop abstract thinking (Hutchison et al., 2011), explore connections between tentative categories and constructs, and facilitate theoretical sorting. Conceptual memos also will elaborate on contradictions and disjuncture and will highlight gaps in the analysis and further data requirements and thus, lead to the theoretical sampling.

Memos continued to be written during the final synthesis of the theoretical model which led to the start of narrative writing. Previous literature will be reviewed around categories and core categories to explore how the findings will be compared to the work of others and to weave the literature within the study findings. Writing and rewriting of this thesis will be utilized to sharpen thinking, clarify ideas, and advance the analysis to formulate the theory and underlying concepts in an abstract fashion. These practices are also recommended by Charmaz (2008) to ensure that the theory's grounding is substantial.

Finally, the theory to be formulated for this study will be grounded within the data from which it is derived and is considered to be substantive theory, as it is closely linked with the empirical situation and will link to practice and interaction (Charmaz, 2006). The theory is not a formal theory as proposed by Glaser (2007) as application and generalization beyond emerging adult students of Capiz is not proposed. The theory to be constructed may however be able to explain similar situations.

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(14) Major Activities/Work Plan

Activities	Year 1										
1. Data Collection 2. Analysis of Data 3. Writing of Research Report 4. Presentation & Publication	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec

(15) Expected Output

Published journal article of the research.

(16) Technology Transfer Activity Design

(17) Targets Beneficiaries

The primary beneficiaries of this study are Filipino learners engaged in mathematics education within the diverse educational settings of the Philippines. The study specifically aims to shed light on the development and manifestation of mathematical grit among these students. By delving into the multifaceted nature of grit perceived by Filipino learners, exploring the factors influencing grit development, and understanding how challenges in mathematical learning shape grit cultivation, the research directly benefits the target population.

Furthermore, the insights garnered from this study are poised to contribute significantly to the broader educational landscape in the Philippines. The development of a comprehensive theory of mathematical grit has the potential to inform educational policies, curriculum development, and instructional strategies tailored to the unique cultural context and challenges encountered by Filipino students in mathematics learning. As such, educators, policymakers, and curriculum developers stand to gain valuable insights from the study's outcomes, contributing to the enhancement of educational practices and the overall learning experience for Filipino students. Additionally, the research findings may have implications for educational researchers seeking to further understand and explore the nuances of grit in diverse cultural and educational contexts.

(18) Personnel Requirement

Name	Role	Field of Specialization
1. Alvin D. Tenorio	Study Leader	Mathematics, Educational Research
2. Leizyl Igdañes	Co-researcher	Mathematics
3. John Patrick Eswan	Co-researcher	Mathematics
Percent Time Devoted to the Program/ Project/ Study		

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Program/Project Leader (in hours):

Program/Project Members (in hours):

(19) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (Php)
A. Personnel Services (For studies needing Research Assistants, Informant/s, Guide)	1 RA for 2 months	P22,000.00
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		P10,000.00
- Meal allowance during data gathering		P10,000.00
- Professional fees (laboratory analysis and other related fees)		
- Miscellaneous (for incidental expenses)		P5,000.00
- Presentation		P10,000.00
- Publication		P30,000.00
Total		P87,000.00

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